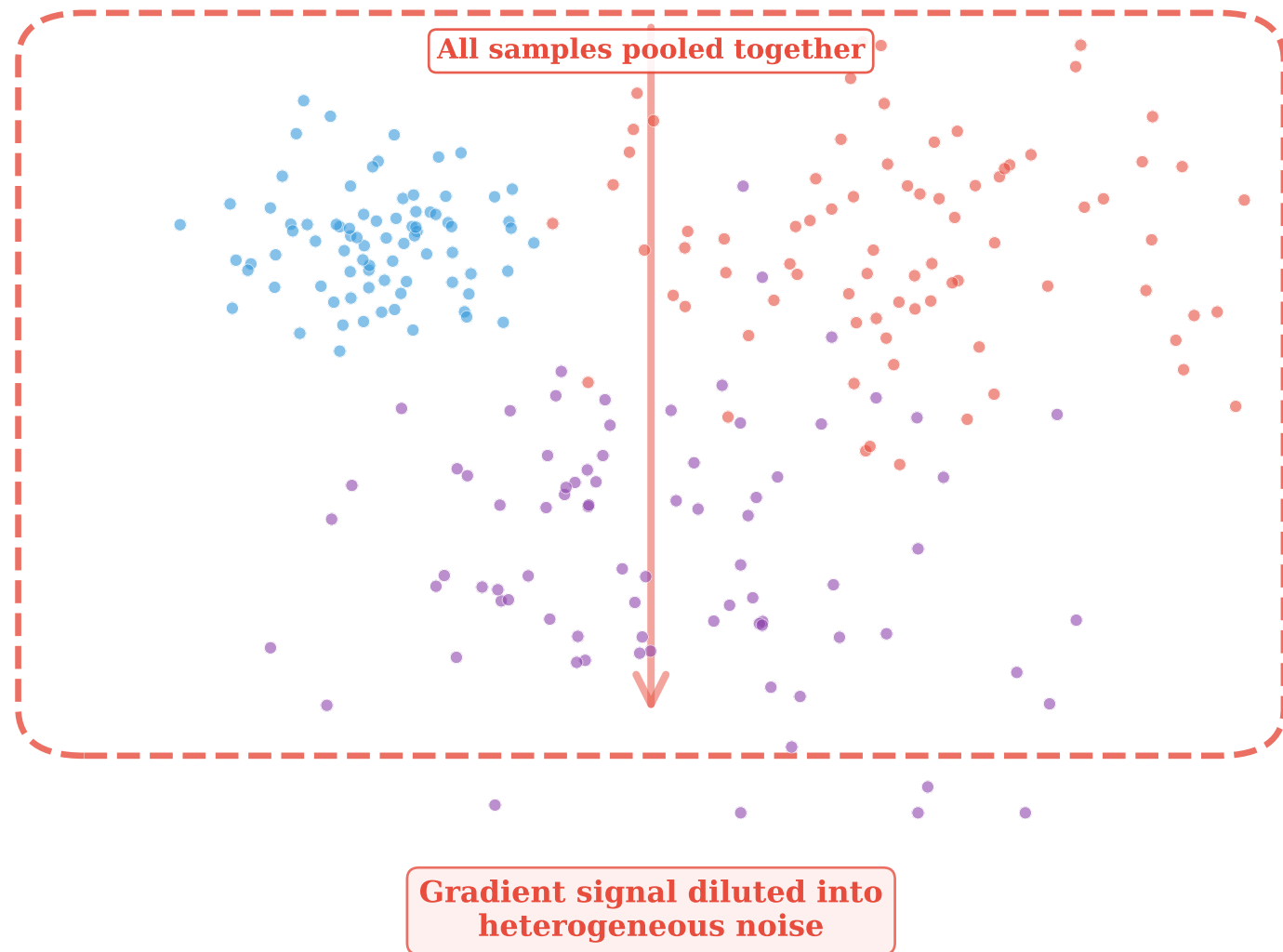


(a)

NOTEARS: Pooled Optimization

(b)

CAGate: Cluster-Aware Gating

MAD-aware Gate Weights: $\gamma_k = 1 - \exp(-\alpha \cdot \overline{\text{MAD}}_k)$

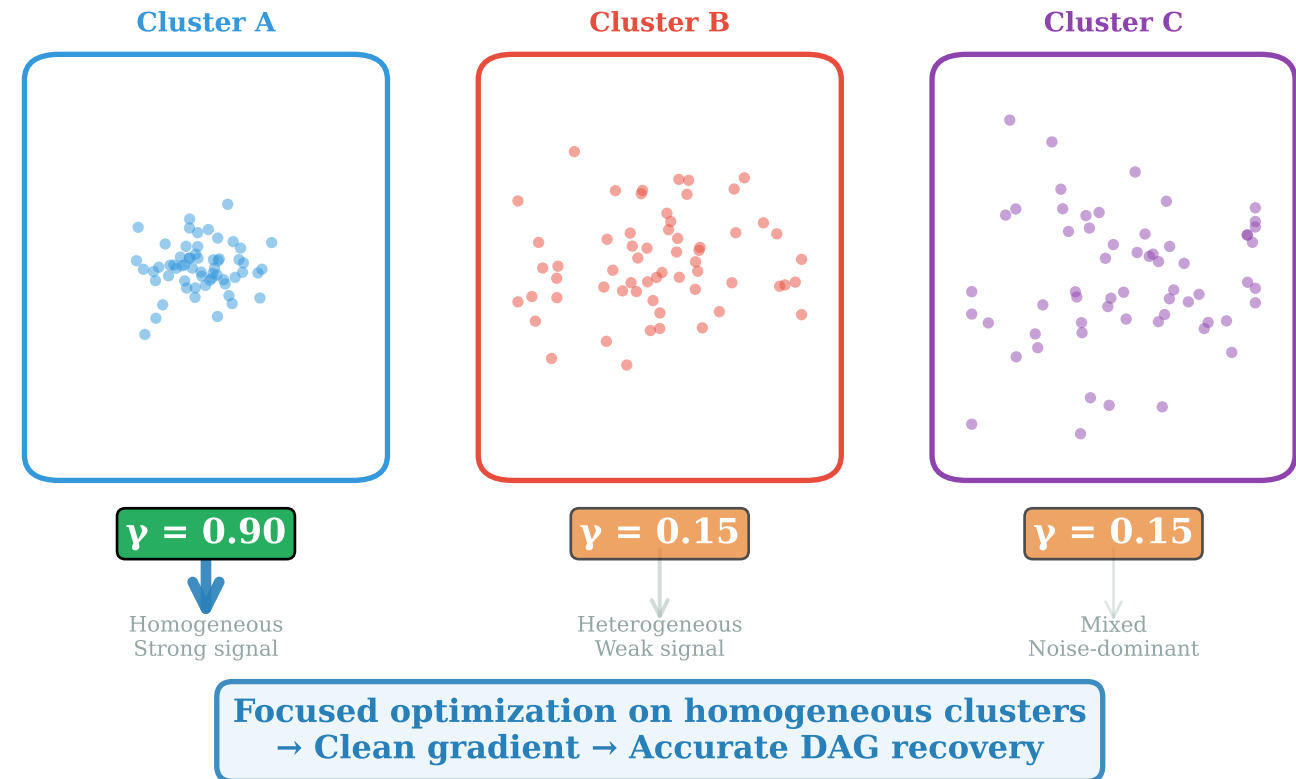


Figure 1. CAGate mechanism: (a) NOTEARS pools all samples, diluting gradient into noise; (b) CAGate partitions samples and applies MAD-aware gating.